KOSHY MAMMOOTTILL MATHEW

DATA SCIENCE | ARTIFICIAL INTELLIGENCE & MACHINE LEARNING | GENERATIVE AI | FORWARD DEPLOYED ENGINEERING (FDE) | DATA ANALYTICS | AUTOMATION

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PROFESSIONAL SUMMARY

Technology entrepreneur and independent developer focused on Data Science, Artificial Intelligence, Machine Learning, Generative AI, Python development, data analytics, automation, and digital product engineering. Develops practical technology solutions through independent projects and a sole proprietorship, combining AI-assisted application development, business process analysis, data-driven problem-solving, software prototyping, and privacy-conscious design.

Hands-on project experience includes developing SignBridgeAI, a continuous American Sign Language (ASL) fingerspelling recognition system using deep learning, and building BusinessDesk by KMM and ResumeFill AI, browser-based digital products designed to address business operations and productivity challenges. Achieved a best validation Character Error Rate (CER) of 16.32% and a later cached FSBoard test-split CER of 15.96% in SignBridgeAI.

Combines a multidisciplinary background in technology, business management, human resources, psychology, and content operations. Interested in Forward Deployed Engineering (FDE), where technical implementation, customer requirements, rapid prototyping, system integration, and iterative solution delivery converge to solve real-world business problems.

CORE COMPETENCIES

- Data Science, Data Analytics and Exploratory Data Analysis
- Artificial Intelligence, Machine Learning and Deep Learning
- Generative AI, Large Language Models (LLMs) and Prompt Engineering
- Computer Vision, Natural Language Processing and Sequence Modelling
- Forward Deployed Engineering (FDE) and Customer-Centric Solution Engineering
- Python Development, Automation and API Integration
- AI-Assisted Software Development and Rapid Prototyping
- Data Processing, ETL/ELT Concepts and Data Engineering
- Business Applications, Digital Product Development and Workflow Automation
- Business Analysis, Requirements Engineering and Process Improvement
- Privacy-Conscious Software Design and Local-First Applications
- Cloud Technologies, DevOps, MLOps and Application Deployment Concepts
- Business Operations, CRM, Reporting and Digital Business Development
- SAP Enterprise Applications, ERP Modules and Cross-Functional Business Process Integration

PROFESSIONAL EXPERIENCE

Business Owner – Proprietor

Koshy Mammoottill Mathew – Sole Proprietorship
Independent Technology Projects and Digital Solutions | September 2026 – Present

Established an independent technology and digital solutions practice focused on AI-assisted development, data analytics, automation, business applications, digital products, and practical software solutions. Manage project planning, requirements analysis, technical research, development, testing, documentation, portfolio development, and business operations.

Key Responsibilities and Technical Contributions

1. AI, Machine Learning and Deep Learning

- Designed and developed SignBridgeAI, a continuous ASL fingerspelling recognition project using Python, TensorFlow/Keras, MediaPipe, Temporal 1D Convolutional Neural Networks, bidirectional GRU layers, and Connectionist Temporal Classification (CTC).
- Implemented hand-landmark extraction, sequence preprocessing, feature normalization, data augmentation, model training, decoding, and Character Error Rate (CER) evaluation.
- Built an interactive Gradio webcam prototype to demonstrate near-real-time recognition and evaluate the practical application of deep learning to computer vision.
- Applied machine learning concepts, model experimentation, evaluation metrics, and iterative refinement to independent analytical and AI development workflows.

2. Data Science and Data Analytics

- Applied Python, Pandas, NumPy, SQL concepts, and Scikit-learn in data processing, exploratory analysis, feature preparation, experimentation, and evaluation workflows.
- Used structured analytical approaches to investigate datasets, identify patterns, prepare features, evaluate outputs, and support evidence-based technical decisions.
- Applied data management and reporting concepts to business applications involving sales, finance, inventory, projects, and operational records.
- Explored analytical methods and scalable data processing approaches relevant to ETL/ELT workflows, data pipelines, data lakes, and business intelligence.

3. Generative AI, LLM Applications and AI-Assisted Development

- Used generative AI tools to accelerate software prototyping, technical research, debugging, implementation planning, documentation, and application development.
- Explored prompt engineering, large language models, Retrieval-Augmented Generation (RAG), semantic search, vector databases, and document retrieval approaches for knowledge-based applications.
- Evaluated opportunities to integrate AI capabilities into business workflows, productivity tools, document processing, and digital products.
- Applied iterative AI-assisted development practices to translate ideas into working prototypes and refine features against practical requirements.

4. Forward Deployed Engineering (FDE) and Solution Engineering

- Applied a solution-oriented development approach by identifying operational needs, translating business problems into technical requirements, and building application features around specific use cases.

- Connected business workflows with software capabilities through rapid prototyping, iterative implementation, functional validation, and requirements refinement.
- Developed practical solutions spanning customer relationship management, sales tracking, invoicing, finance records, inventory, project monitoring, document processing, and reporting.
- Considered application integration, data handling, usability, maintainability, and privacy requirements when designing digital solutions.
- Used feedback-driven and iterative development practices to refine prototypes and align technical implementation with intended business outcomes.
- Evaluated opportunities for API-based integrations, AI-enabled workflows, automation, and deployment-oriented engineering to address operational challenges.

5. Python Development, Software Engineering and Automation

- Developed Python-based workflows and application prototypes using modular programming, object-oriented programming principles, data structures, API integration concepts, and automation techniques.
- Applied software design, debugging, reusable components, and iterative development practices to independent technology projects.
- Explored Flask, Django, and FastAPI for application backends and RESTful service development.
- Studied and applied software architecture concepts relevant to asynchronous programming, microservices, distributed systems, GraphQL, and maintainable application design.
- Evaluated opportunities to automate repetitive business processes and improve the efficiency of data handling, reporting, and digital workflows.

6. Web Applications and Digital Product Development

- Created BusinessDesk by KMM, a browser-based, local-first business management workspace covering CRM, sales opportunities, quotations, invoicing, finance and payment tracking, project management, inventory, team and goal management, reporting, and local backup and restore.
- Developed ResumeFill AI, a browser-based resume-processing and form-filling tool featuring editable profile information, JSON/CSV export, and browser extension-assisted form filling.
- Applied JavaScript, browser-based storage, web application design, and AI-assisted development to create functional digital products.
- Prioritized usability, practical functionality, local data handling, and reduced dependence on centralized cloud storage where appropriate.

7. Data Engineering, Databases and AI Integration

- Applied structured data modelling and SQL-related concepts to business application workflows, data organization, and reporting requirements.
- Worked with data management concepts relevant to relational databases, document databases, caching, and vector-based information retrieval.
- Explored PostgreSQL, MySQL, SQL Server, MongoDB, Redis, SQLAlchemy, Chroma, Pinecone, FAISS, and cloud data lake technologies for application and AI use cases.
- Studied ETL/ELT pipelines, Apache Spark, Hadoop, Kafka, and Airflow to understand data ingestion, transformation, orchestration, and scalable processing.
- Investigated integration patterns for connecting application services, machine learning models, and AI capabilities through APIs and retrieval-based architectures.

8. SAP Modules and Enterprise Business Process Solutions

- Developed familiarity with SAP enterprise application modules and their relevance to business operations, enterprise resource planning (ERP), and integrated business workflows.
- Explored SAP Sales and Distribution (SD), Materials Management (MM), Financial Accounting and Controlling (FICO), Warehouse Management (WM), Production Planning (PP), and Plant

Maintenance (PM) in the context of sales, procurement, finance, inventory, manufacturing, and maintenance processes.
- Studied SAP BASIS, Process Integration/Process Orchestration (PI/PO), ABAP development, and SAP Security/Governance, Risk, and Compliance (GRC) concepts relevant to system administration, enterprise integration, customization, access control, and compliance.
- Evaluated opportunities to align SAP-related business processes with data analytics, reporting, workflow automation, API integration, and AI-enabled enterprise solutions.
- Applied a solution-oriented perspective to understanding business requirements, cross-functional process dependencies, enterprise application integration, and digital transformation opportunities.

9. Cloud, DevOps, MLOps and Deployment

- Explored application deployment, model-serving concepts, version control, reproducible development workflows, and CI/CD principles.
- Studied AWS, Microsoft Azure, Google Cloud Platform, Docker, Kubernetes, Helm, Terraform, Ansible, and infrastructure automation concepts.
- Investigated MLflow, CI/CD for machine learning pipelines, GitHub Actions, GitLab CI, Jenkins, monitoring, and model lifecycle management.
- Developed an understanding of deployment architecture, environment management, scalability, and operational considerations for software and AI applications.

10. Software Testing and Quality Assurance

- Applied iterative testing, debugging, functional validation, and feature verification throughout independent application development.
- Explored automated testing, API testing, browser automation, and performance testing approaches using pytest, Selenium, Cypress, Playwright, Postman, JUnit, TestNG, and related tools.
- Considered reliability, correctness, usability, and maintainability when refining software prototypes and digital products.

11. Business Analysis, Operations and Process Improvement

- Translated business requirements into application features, functional workflows, and implementation priorities.
- Designed business application workflows for CRM, sales opportunity tracking, quotations, invoicing, payment tracking, inventory, project management, goals, and reporting.
- Applied requirements analysis, process mapping, documentation, prioritization, project planning, and iterative delivery principles.
- Used business process improvement concepts to identify opportunities for operational efficiency, workflow automation, and integrated digital solutions.

12. Business Development, CRM and Client Solutions

- Defined independent digital service offerings around AI-assisted development, Python automation, data analytics, reporting, web applications, and business process improvement.
- Explored B2B and B2C business models, lead generation, customer acquisition, CRM practices, stakeholder communication, and online business development.
- Evaluated how digital solutions could address customer needs, simplify operations, and support small-business administration.
- Developed portfolio assets and project demonstrations to communicate technical capabilities and potential business value.

13. Digital Marketing, SEO and Online Presence

- Applied digital positioning, content planning, portfolio presentation, and website development principles to independent business and technology projects.
- Studied SEO, technical SEO, Google Analytics 4, Google Ads, Meta Ads Manager, LinkedIn Ads, SEMrush, Ahrefs, social media strategy, and performance marketing.
- Explored digital content creation, creative writing, visual communication, and online outreach to support service visibility and customer acquisition.

14. Project Management and Independent Business Operations

- Manage independent project planning, technical research, feature development, documentation, iterative refinement, and portfolio maintenance.
- Coordinate the technical and operational aspects of digital product development, from identifying a problem to defining requirements and building a working prototype.
- Evaluate emerging technologies and business opportunities in AI, automation, data analytics, and digital product engineering.
- Apply business planning, reporting, research, documentation, and workflow improvement principles to the development of independent technology solutions.

Content Reviewer – Trust Operations

Amazon Ads | July 2022 – April 2025

- Reviewed and evaluated advertising content against established policies, content guidelines, and quality standards.
- Processed high volumes of cases while maintaining accuracy, productivity, and adherence to operational requirements.
- Applied policy guidelines consistently and escalated complex or ambiguous cases when necessary.
- Followed established workflows and performance expectations to maintain content quality and operational consistency.
- Collaborated with team members to uphold quality standards, resolve operational issues, and support continuous process improvement.
- Developed experience in structured decision-making, policy interpretation, quality assurance, and process-driven operations within a large-scale digital advertising environment.

SELECTED PROJECTS

1. SignBridgeAI – Continuous ASL Fingerspelling Recognition Using Deep Learning

Objective: Develop a computer vision and deep learning system that recognizes continuous American Sign Language fingerspelling and converts it into text.

Technical Implementation

- Built a hand-landmark-based recognition pipeline using MediaPipe, extracting 63 landmark features per frame.
- Implemented video and sequence preprocessing, normalization, data augmentation, and training data preparation.
- Developed a hybrid temporal deep learning architecture combining Temporal 1D Convolutional Neural Networks and bidirectional GRU layers.
- Used Connectionist Temporal Classification (CTC) for sequence-level decoding without requiring separate character-level alignment for every frame.
- Implemented model training, decoding, and Character Error Rate (CER) evaluation using TensorFlow/Keras.
- Developed an interactive Gradio webcam prototype for near-real-time recognition.

Results

- Best validation Character Error Rate: **16.32%**
- Later cached FSBoard test-split Character Error Rate: **15.96%**

Technologies: Python, TensorFlow, Keras, MediaPipe, OpenCV, NumPy, Pandas, Gradio, Google Colab.

GitHub: https://github.com/KoshyMammoottillMathew/SignBridgeAI

2. BusinessDesk by KMM – Local-First Business Management Workspace

Objective: Develop a privacy-conscious, browser-based workspace to organize small-business operations without depending on a centralized application cloud database.

Key Features

- Customer relationship management and sales opportunity tracking.
- Quotation preparation, invoicing, finance records, and payment tracking.
- Project, inventory, team, and goal management.
- Business reporting and local backup and restore functionality.
- Browser-based local data storage designed to keep business information under the user's control.

Technologies: JavaScript, web technologies, browser-based storage, Git, GitHub Pages, AI-assisted development.

Project: https://koshymammoottillmathew.github.io/products/businessdesk-kmm/

3. ResumeFill AI – Privacy-Focused Resume Processing and Form-Filling Tool

Objective: Simplify resume data extraction and job application form completion while prioritizing local data handling and user control.

Key Features

- Resume PDF data extraction and editable profile information.
- JSON and CSV export.
- Browser extension-assisted form filling.
- Local-first design intended to reduce unnecessary data sharing and dependence on cloud storage.

Technologies: JavaScript, web technologies, PDF processing, browser extension development, local data handling, GitHub Pages, AI-assisted development.

Project: https://koshymammoottillmathew.github.io/products/resumefill-ai/

TECHNICAL SKILLS

Programming Languages: Python, R, C++, Java, C#, Scala, PHP, JavaScript, SQL, XML, T-SQL.

Data Science, AI and Machine Learning: Data preprocessing, exploratory data analysis, feature engineering concepts, machine learning, deep learning, computer vision, natural language processing (NLP), Generative AI, large language models (LLMs), prompt engineering, Retrieval-Augmented Generation (RAG), sequence modelling, CNNs, GRUs, CTC decoding, Hugging Face, OpenCV, TensorFlow, Keras, PyTorch, Scikit-learn, AI ethics and governance, enterprise AI architecture, LLM integration, AI risk assessment, AI solution design.

Forward Deployed Engineering and Solution Delivery: Customer-centric problem-solving, business requirements analysis, rapid prototyping, solution engineering, workflow integration, API integration concepts, technical discovery, iterative development, functional validation, deployment-oriented thinking, stakeholder communication, requirements refinement, feedback-driven improvement.

Data Engineering and Databases: ETL/ELT pipelines, data processing, Apache Spark, Hadoop, Kafka, Airflow, PostgreSQL, MySQL, SQL Server, MongoDB, Snowflake, BigQuery, Redis, SQLAlchemy, vector databases, Chroma, Pinecone, FAISS, cloud data lakes.

SAP and Enterprise Resource Planning (ERP): SAP SD (Sales and Distribution), SAP MM (Materials Management), SAP FICO (Financial Accounting and Controlling), SAP WM (Warehouse Management), SAP PP (Production Planning), SAP PM (Plant Maintenance), SAP BASIS (System Administration), SAP PI/PO (Process Integration/Process Orchestration), SAP ABAP (Advanced Business Application Programming), SAP Security/GRC (Governance, Risk, and Compliance), enterprise business processes, ERP concepts, cross-functional workflow integration, business process analysis, enterprise application integration.

Python Development and Software Architecture: Advanced Python, object-oriented programming, data structures and algorithms, Django, FastAPI, Flask, RESTful APIs, GraphQL APIs, REST API integration, microservices, system design, distributed systems, asynchronous programming, AsyncIO, Celery.

Java and Full-Stack Development: Java 11+, Spring Boot, Spring Cloud, Hibernate, JPA, React.js, Angular, Next.js.

.NET and Application Development: .NET Core, .NET 6+, ASP.NET Web API, Entity Framework Core, SQL Server, Blazor, Laravel, Odoo Framework, Odoo custom module development.

Cloud, DevOps and MLOps: AWS, Amazon EC2, Amazon EKS, Amazon S3, AWS IAM, Amazon CloudFront, AWS Lambda, Microsoft Azure, Google Cloud Platform, Docker, Kubernetes, Helm, Terraform, Ansible, AWS CloudFormation, CI/CD pipelines, CI/CD for ML pipelines, MLflow, Jenkins, GitHub Actions, GitLab CI, Prometheus, Grafana, ELK Stack, model deployment concepts.

Software Testing and Quality Assurance: Selenium, Cypress, Playwright, Appium, pytest, TestNG, JUnit, Postman, REST Assured, JMeter, Locust, LoadRunner, Behaviour-Driven Development (BDD), Cucumber, automated testing, API testing, performance testing, debugging, functional validation.

Business Analysis and Project Management: Agile, Scrum, Kanban, Waterfall, JIRA, Confluence, Trello, Microsoft Visio, Lucidchart, requirements gathering, Business Requirements Documents (BRD), Functional Requirements Documents (FRD), user stories, process mapping, stakeholder management, risk management, business case formulation, solution design, project planning.

Business Development and CRM: Lead generation, cold calling, B2B sales, B2C sales, client relationship management, Salesforce, HubSpot, Zoho CRM, LinkedIn Sales Navigator, stakeholder communication, customer acquisition.

Digital Marketing and Content: Digital marketing, performance marketing, Google Ads, Meta Ads Manager, LinkedIn Ads, search engine optimization (SEO), technical SEO, Google Analytics 4 (GA4), SEMrush, Ahrefs, content strategy, creative writing, social media strategy, Canva, Adobe Photoshop, Adobe Premiere Pro, Adobe Illustrator.

Business Operations and Digital Solutions: Business planning, workflow automation, business process improvement, CRM solutions, sales tracking, invoicing, project management, reporting, research, documentation, digital product development, AI-assisted development, prototyping, online business development, operational efficiency.

EDUCATION AND CERTIFICATIONS

Certificate Course in Data Science & AI

KELTRON | 2026

Completed training in Data Science and Artificial Intelligence.

Master of Business Administration (MBA) – Human Resource Management

Indira Gandhi National Open University (IGNOU) | 2025

Academic project: Emotional Intelligence in Performance Management, with research focused on NABARD.

Master of Arts (MA) – Clinical Psychology

Indira Gandhi National Open University (IGNOU) | 2021

Research: Happiness and Personality Traits among Adolescent Students.

Clinical exposure: Internship at General Hospital, Thiruvananthapuram.

Bachelor of Arts (BA) – English Literature

Indira Gandhi National Open University (IGNOU) | 2017

ADDITIONAL PROFESSIONAL STRENGTHS

- Analytical thinking and structured problem-solving.
- Independent project ownership and self-directed learning.
- Translating business needs into practical technical solutions.
- Technical research, documentation, and iterative development.
- Quality-focused execution and process consistency.
- Cross-functional understanding of technology, business operations, and customer needs.
- Privacy-conscious design and practical application of AI-assisted development.